Multispectral satellite imagery for illegal waste material classification

State of the art and thesis direction — WorldView-3 + Pléiades Neo

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The problem: from detection to risk

Illegal waste dumps: a hidden, widespread environmental crime

Define: EO = Earth Observation · MS = multispectral · GSD = ground sampling distance · VNIR/SWIR = visible-near / short-wave infrared

- Lombardy and Italy face thousands of unauthorised dumping sites — many never reported, only stumbled upon.
- ARPA (the regional environmental agency) must find them and decide where to send a limited number of inspectors first.
- The operational question is not only “where is the waste?” but “how dangerous is it?” — classify by risk, not merely detect.
  - Inspector time is the scarce resource; without ranking, agencies inspect blindly.

Framing: Fraternali et al. 2024 (PoliMi survey, arXiv:2402.09066)

Not all waste is equal: risk = hazard × exposure × magnitude

- The danger is material-bound: inert rubble vs plastics vs asbestos-cement are worlds apart.
- “Material” carries the hazard; the European Waste Catalogue (EWC) flags hazardous codes with “*” — asbestos = 17 06 05*.
- Priority risk = hazard (material) × exposure (who/what is nearby) × magnitude (how much).
  - So material identification is the missing decision variable, not just site detection.

EWC List of Waste (Dec. 2000/532/EC) · Indice di Degrado d.d.g. 13237/2008 · Fazzo et al. 2023

From pixels to an ARPA intervention priority

Indice di Degrado (d.d.g. 13237/2008) is NOT in the public WFS - estimated remotely (SWIR Mg-OH + VNIR weathering) = a thesis contribution

EWC List of Waste 2000/532/EC . Indice di Degrado d.d.g. 13237/2008 . Fazzo et al. 2023

The research question

- What is the added value of multispectral (MS) data, vs RGB only, for waste MATERIAL classification from satellite imagery?
  - Measured, not assumed — and explicitly bounded: where does multiband help, and where does it stop?
  - Two confirmed very-high-resolution sensors: WorldView-3 (with SWIR) and Pléiades Neo (VNIR-only).

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Why ordinary RGB is not enough

Today’s paradigm: RGB deep learning detects sites, not materials

- Pipeline: VHR RGB tile → CNN/Transformer backbone → EO pretraining → two-step fine-tune → binary waste / no-waste.
- Strong within the geographic context it was trained on: Gibellini et al. 2025 — F1 92.0 %, Acc 94.6 % (20 cm, Swin-T + RSP).
- But three limits motivate the spectrum:
  - classifies presence, not material;
  - tied to colour only (3 broad bands);
  - generalizzazione collapses cross-region (−5.1 % F1: GR 85.5 / SE 83.8 / RO 91.5) and cross-sensor.

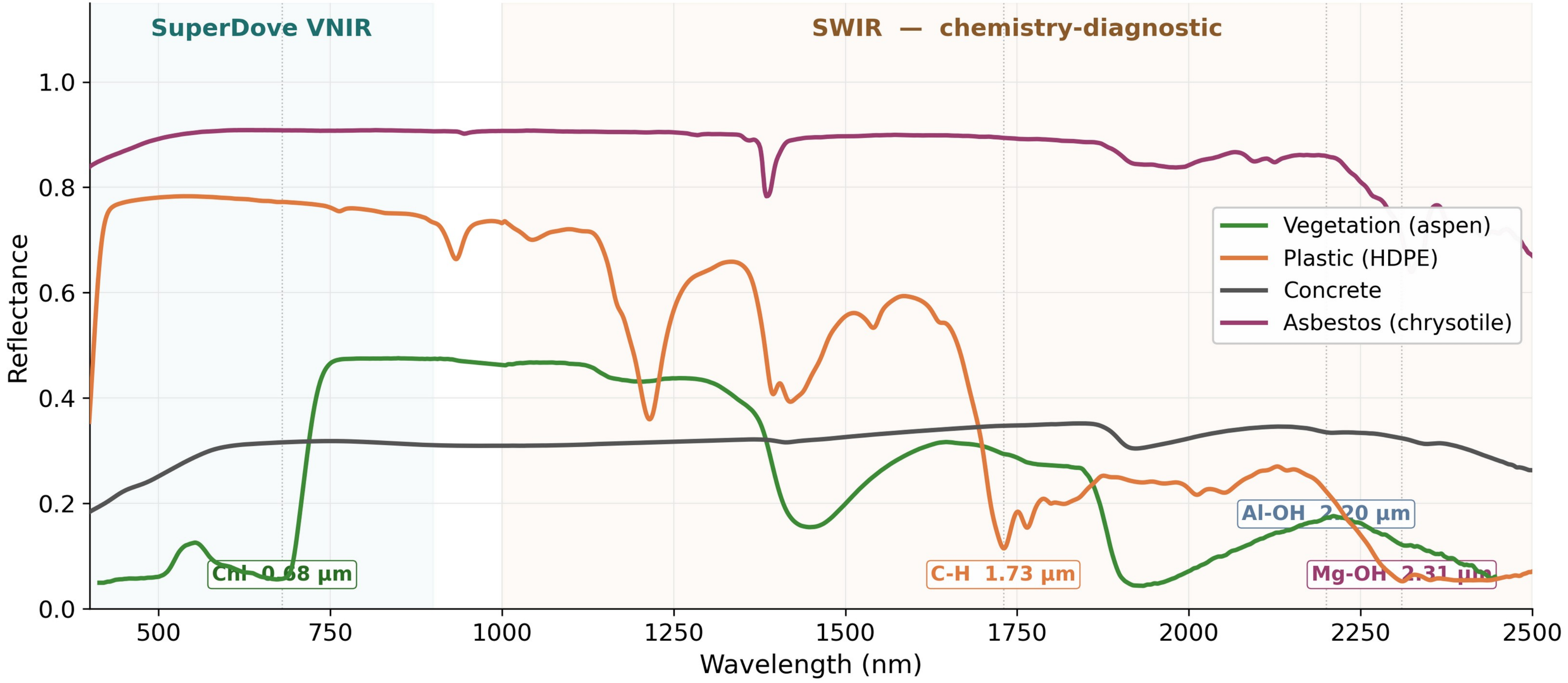
Gibellini et al. 2025 (Waste Mgmt Bull., arXiv:2502.06607) · Fraternali et al. 2024

A satellite pixel is a spectrum, not just a colour

- Each pixel is a vector of reflectance values — the material's spectral signature.
- RGB: 3 broad visible bands (colour only).
- MS (multispectral): 4–15 chosen bands. WorldView-3 records 8 VNIR + 8 SWIR; Pléiades Neo 6 VNIR.
- HSI (hyperspectral): hundreds of contiguous narrow bands (e.g. EnMAP 230, splib07a 2,151 channels @ 350–2500 nm).
  - This vector is the raw input from which any classifier reasons.

Spectrum schema after USGS splib07a (Kokaly et al., 2017)

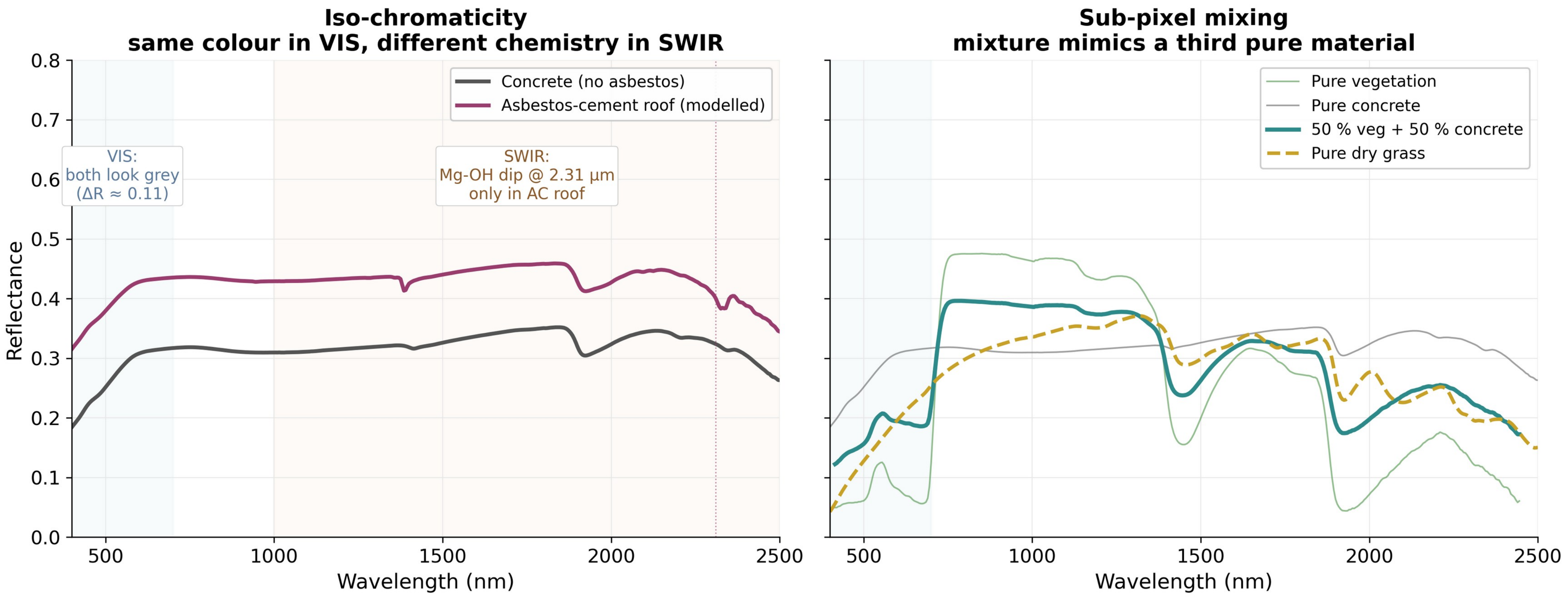
Every material has a spectral fingerprint



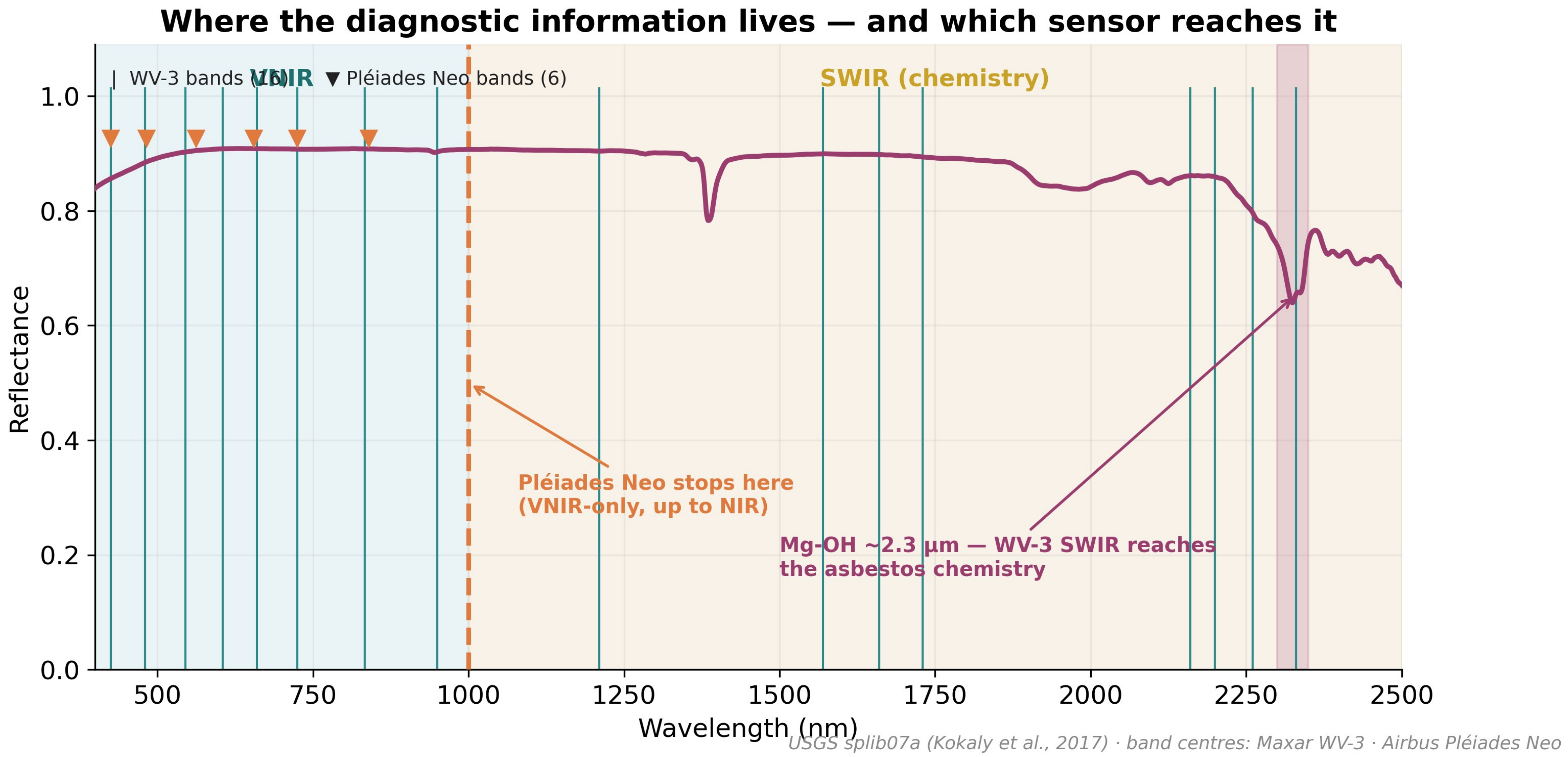
USGS splib07a (Kokaly et al., 2017)

Reflectance curves diagnose chemistry, not just colour — USGS splib07a (Kokaly et al., 2017)

RGB fails in two distinct ways



USGS splib07a (Kokaly et al., 2017)



The chemistry-diagnostic clues sit in NIR and especially SWIR — USGS splib07a

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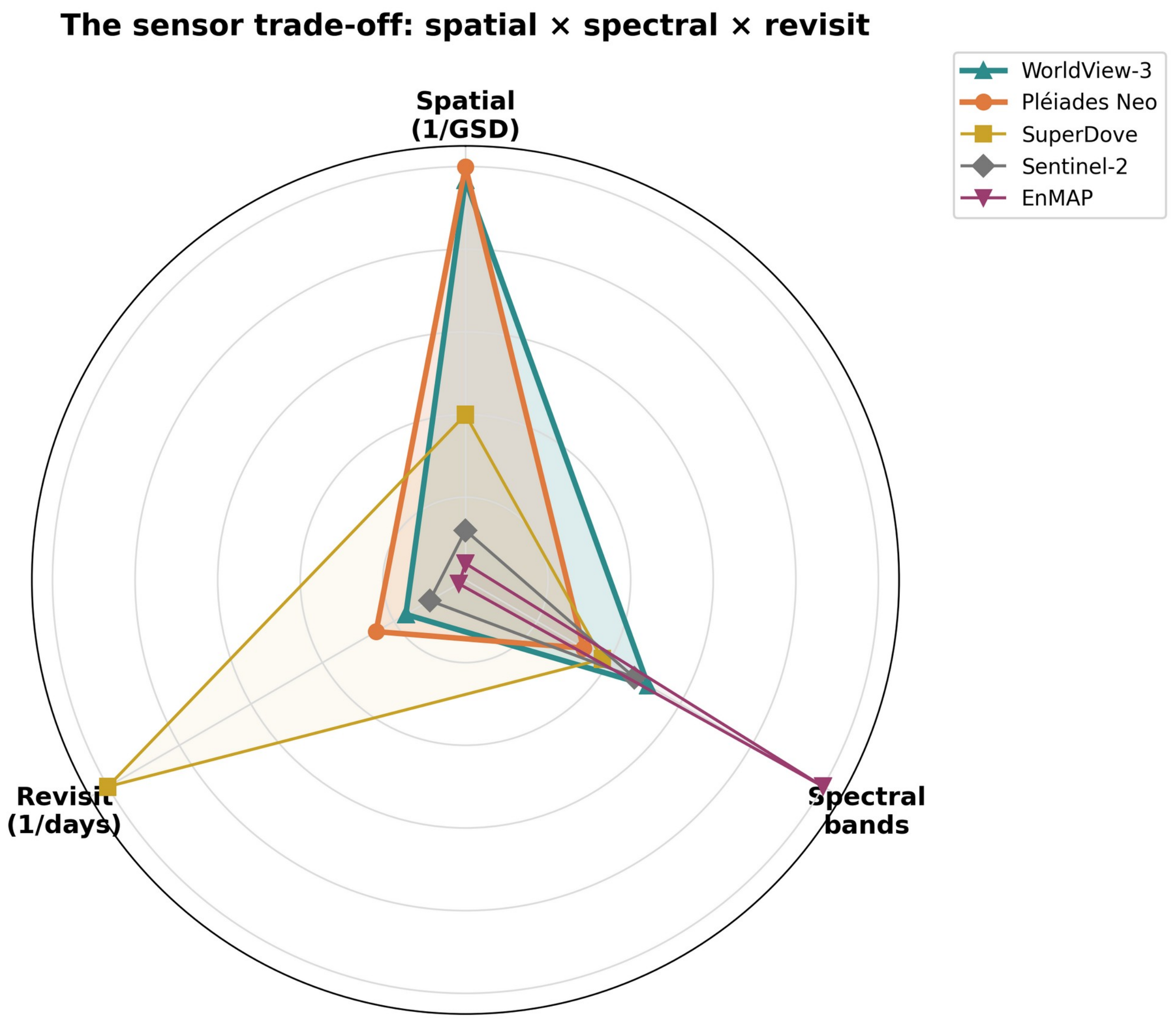
The sensors — what exists today



Outer ring = better. The thesis pair (WV-3, Pléiades Neo) emphasised. Specs: Maxar · Airbus · Planet · ESA · DLR.

Three axes, one photon budget — no satellite maximises all three

Multispectral at low / medium resolution (10–30 m): too coarse for material

Low resolution answers “where”, not “what”

Sensor	Bands	GSD	SWIR	Verdict for material
Sentinel-2	13	10–20 m	20 m	detection only; pixels too mixed
PRISMA / EnMAP	200+ (HSI)	30 m	yes	resolves 2.3 µm triplet but 30 m
Landsat 8/9	11	30 m	yes	regional context only

Kikaki 2022 · Magyar 2023 · Shepherd 2025 · Sun 2023

Low / medium resolution in practice

- MARIDA (Kikaki 2022): marine debris confused with natural organic matter where pixels mix.
- Global dumpsites (Sun 2023, Sentinel-2): 763 dumpsite LOCATIONS across 28 cities (sensitivity 98 %, precision 70 %) — but material identity stays out of reach; mapping in 6 days vs 6 months manual.
- Tisza (Magyar 2023): change detection works; per-tile post-processing still needed.
  - Takeaway: low-res localises sites; it cannot tell what they are made of. → we need VHR.

Kikaki 2022 PLOS ONE · Sun 2023 Nat. Commun. 14 · Magyar 2023

The chosen data: two very-high-resolution satellites

WorldView-3

Maxar · commercial / ESA TPM

VNIR	8 bands @ 1.24 m
SWIR	8 bands @ 3.7 m
Pan	~0.31 m
Unique	Yellow · NIR2
Role	full ladder incl. R3 chemistry

SWIR ✓ rare civilian SWIR at VHR

same sites ↔

Pléiades Neo

Airbus · commercial / ESA TPM

VNIR	6 bands @ 1.2 m
SWIR	none
Pan	~0.30 m
Unique	Deep Blue
Role	VNIR-only cross-sensor axis

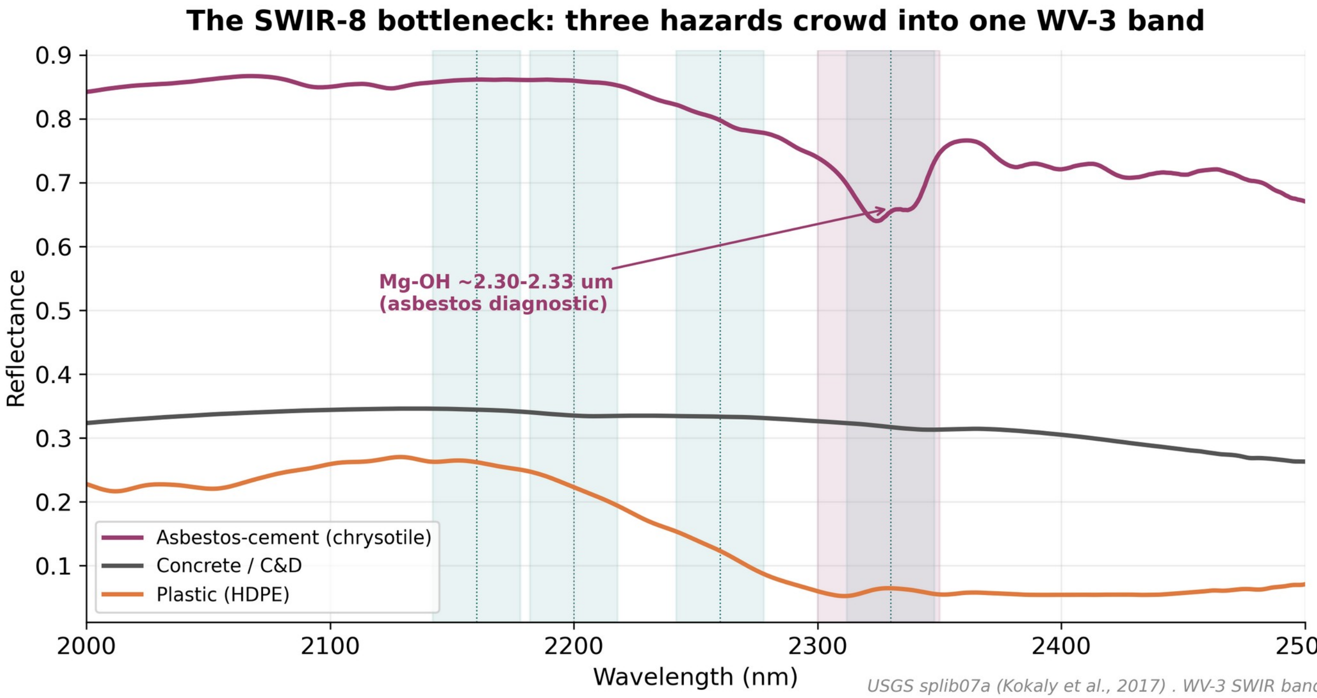
SWIR ✗ tests how far VNIR-only gets

Both sub-metre VHR · access feasible for free via ESA Third-Party Missions (~9-week proposal lead + 1-yr quota)

Specs: Maxar WV-3 · Airbus Pléiades Neo · ESA TPM

Honest caveat: resolution is split between texture and chemistry

- Spectra sampled coarsely: VNIR ~1.24 m, SWIR ~3.7 m (~14 m^2 pixels — no pure pixels over a dump).
- Pan gives fine texture ~0.3 m but no material info alone.
- A small dump is easier to SEE (texture) than to chemically IDENTIFY (spectra).
- SWIR-8 bottleneck: asbestos 2.32 + concrete 2.34 + plastic 2.31 crowd one band.
- Honest: at 3.7 m + atmosphere, R3 may ≈ R2 — itself a valid finding.



USGS splib07a (Kokaly et al., 2017) . WV-3 SWIR band centres: Maxar

asbestos 2.32 . concrete 2.34 . plastic 2.31 - all inside WV-3 SWIR-7 (~2.33 um); discrimination leans on shoulders + VNIR shape

USGS splib07a · WV-3 SWIR band centres (Maxar)

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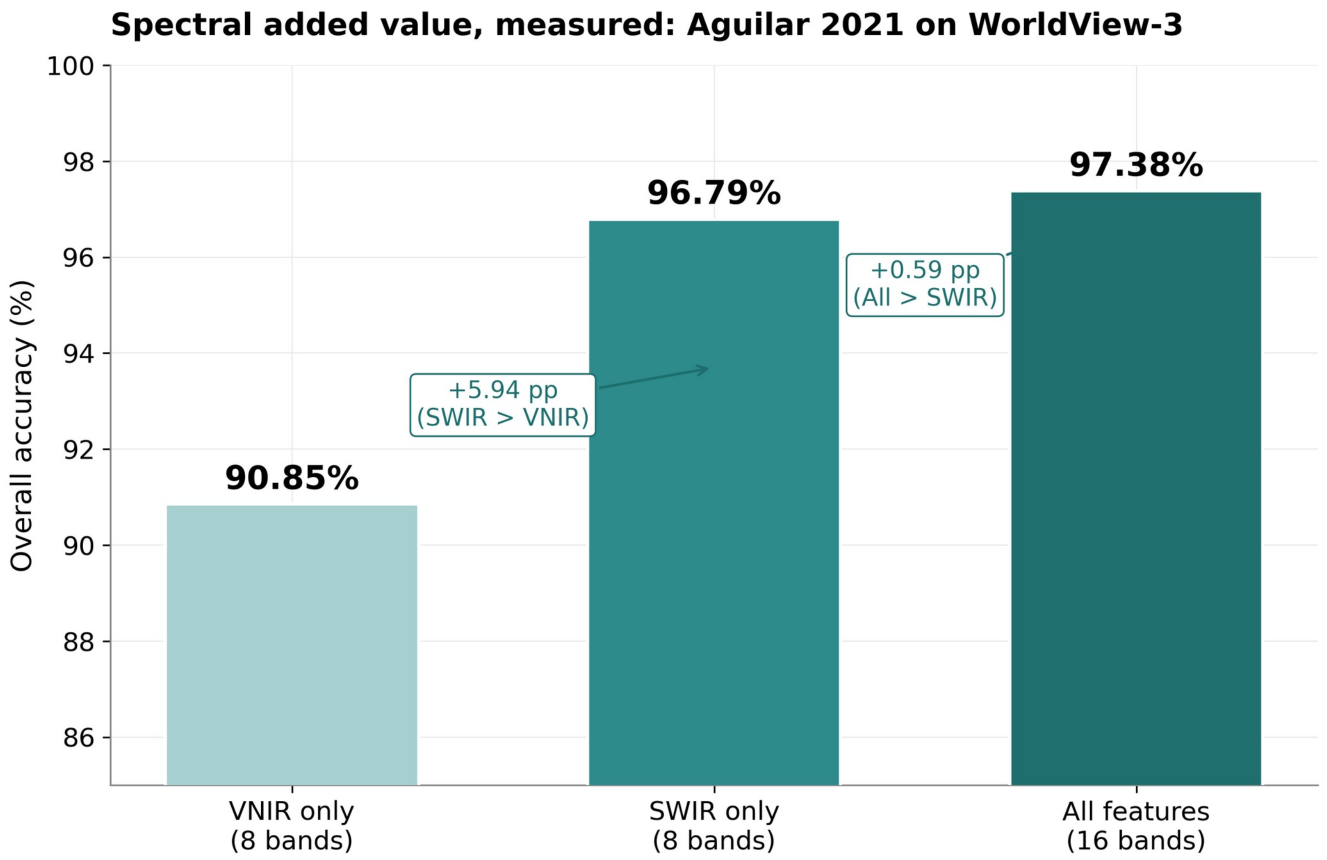
The added value is real — precedents

Spectral added value, measured: Aguilar 2021 on WorldView-3

- WV-3 greenhouse-plastic ablation, OBIA + Decision Tree, 10-fold CV.
- Overall Accuracy:
  - VNIR only → 90.85 %
  - SWIR only → 96.79 %
  - All 16 bands → 97.38 %
- SWIR carries the single largest jump (+6.5 pp over VNIR).
- Most-cited direct benchmark for waste-like material classification.

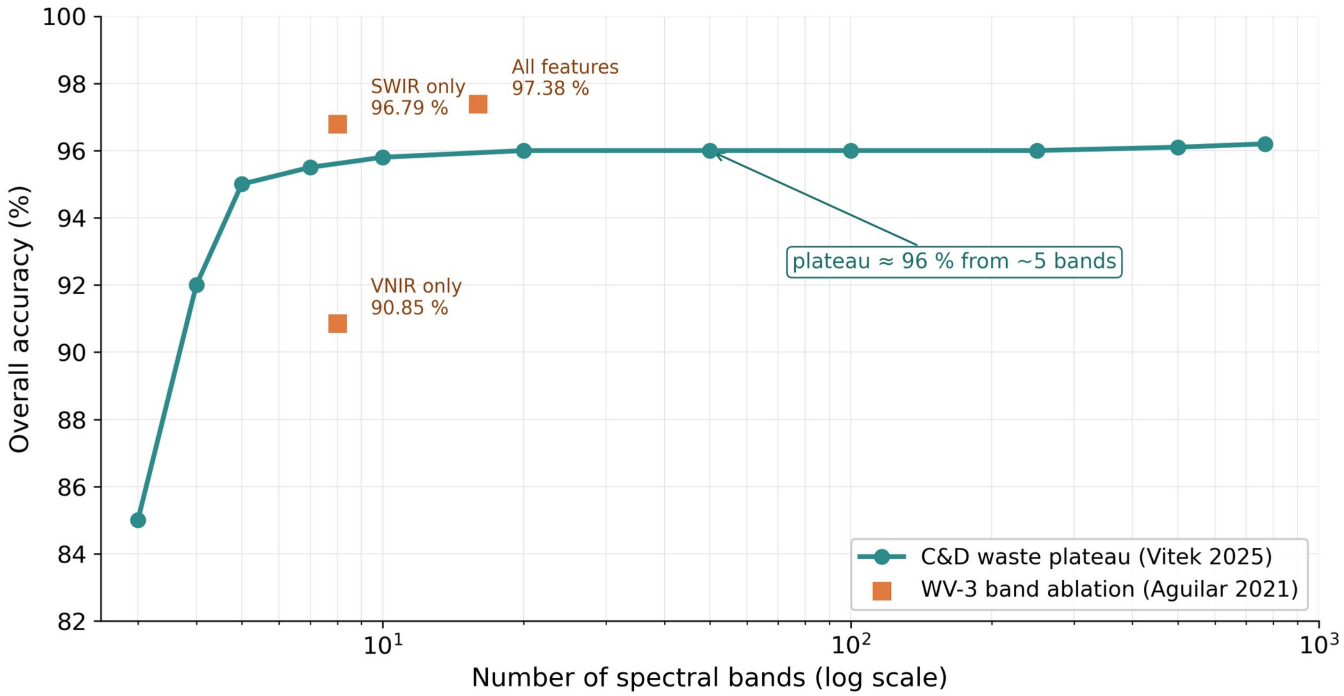


Aguilar, Jiménez-Lao & Aguilar 2021 — Remote Sensing 13(11):2133. OBIA + Decision Tree, 10-fold CV, 14.3 M pixel GT.

Aguilar, Jiménez-Lao & Aguilar 2021, Remote Sensing 13(11):2133 — κ 0.812/0.932/0.944

More bands ≠ more — well-chosen few suffice

- Aguilar 2021 (WV-3 plastic): VNIR 90.85 → All 97.38 — SWIR carries the jump.
- Vitek 2025 (C&D waste): RGB + 2 narrowbands ≈ HSI 768 bands.
- Zhou 2021 (WV-3 plastic): 8 SWIR bands separate 3 polymer clusters.
- What matters is WHICH bands, not how many.
- WV-3 gives us SWIR; Pléiades Neo (VNIR-only) measures how far we get without it.



Vitek et al. 2025 (CTU Prague, C&D waste) — Stage-1/2 narrowband selection on HSI 425–975 nm · Aguilar 2021 verified above.

Vitek et al. 2025 (CTU Prague) · Aguilar et al. 2021 · Zhou et al. 2021

Asbestos precedents — direct comparison points

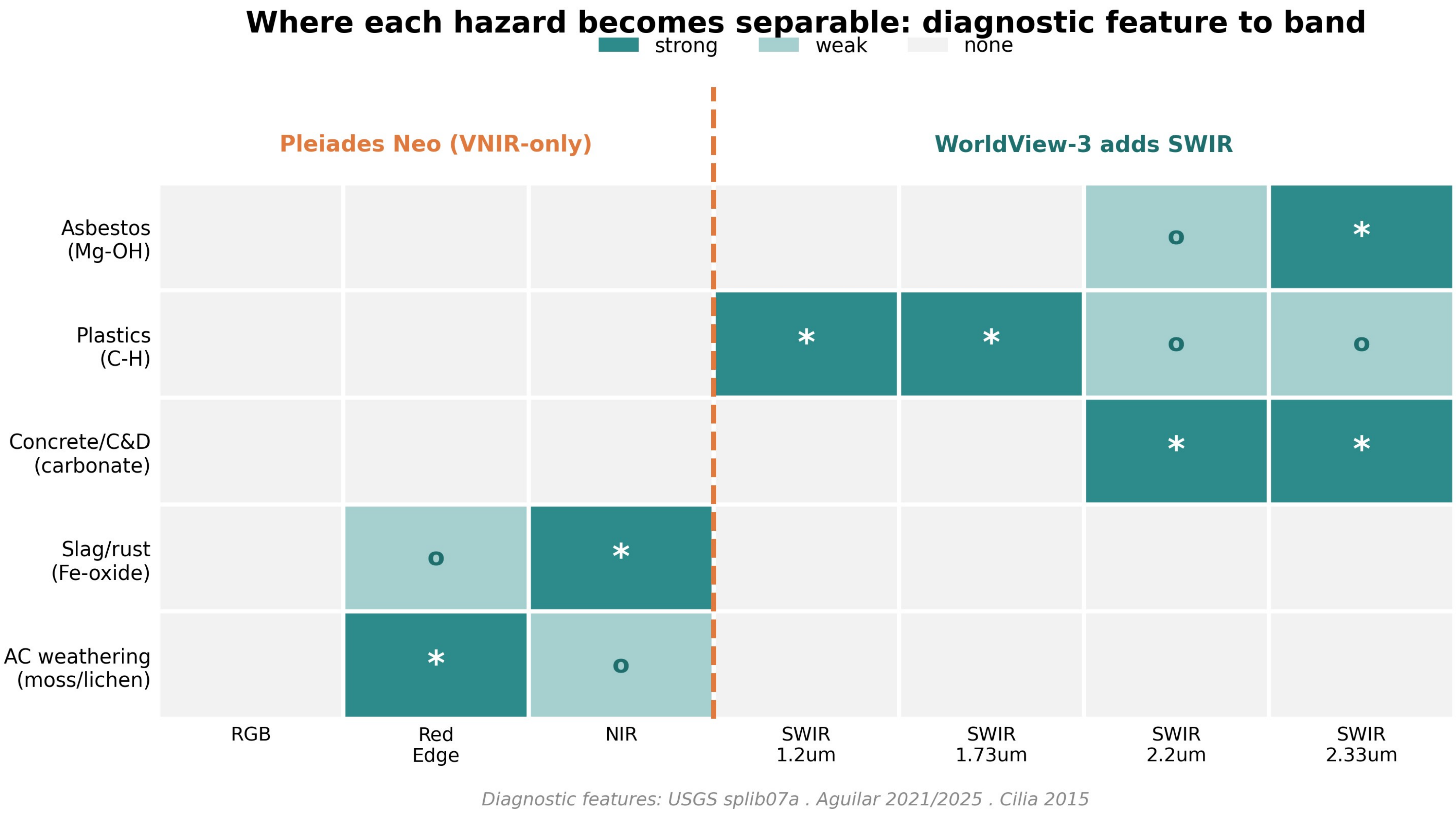
- Saba 2026 (WV-3 VNIR, 8 bands): 32 classifiers tested; Fine-KNN Macro-F1 97.6 % — strong VNIR-only upper bound to beat. [paywalled, single-source]
- Bonifazi 2026 (WV-3, 16 bands, MLC): asbestos-cement roofs F1 0.87 (P 0.81 / R 0.92); 25,319 buildings processed; multi-temporal monitoring.
- Shepherd 2025 (EnMAP, 230 bands @ 30 m): ACE classifier 91.4 % OA / κ 0.87 / F1 91.2 %; 86 % field match — but urban mixed pixels (rubble, paint) mimic asbestos.
  - Together: VNIR can detect AC; SWIR + hyperspectral add robustness on clean/ambiguous roofs.

Saba 2026 J. Hazard. Mater. · Bonifazi 2026 Geomatics 6(3):41 · Shepherd 2025 Sci. Rep.

Waste-dump precedents at VHR

- CascadeDumpNet (Zhang & Ma 2024, Pléiades 0.5 m): open-dumpsite detection, 84.6 % mAP; transfers Shenzhen → Shanghai/Guangzhou.
- AerialWaste (Torres 2023): 11,700+ RGB tiles, 22 material categories — but only 11 images contain presumed asbestos; RGB, object-level.
- Aguilar 2025 (WV-3 SWIR macroplastics): precision 92.5 %, lab-to-satellite r = 0.95; targets 80–150 m^2 aggregations.
  - VHR detection is solved on RGB; material/hazard identification is the open frontier.

Zhang & Ma 2024 RSE · Torres 2023 Sci. Data · Aguilar 2025 Env. Monit. Assess.



Pléiades Neo reaches up to NIR; WorldView-3 adds the SWIR chemistry — USGS splib07a · Aguilar · Cilia 2015

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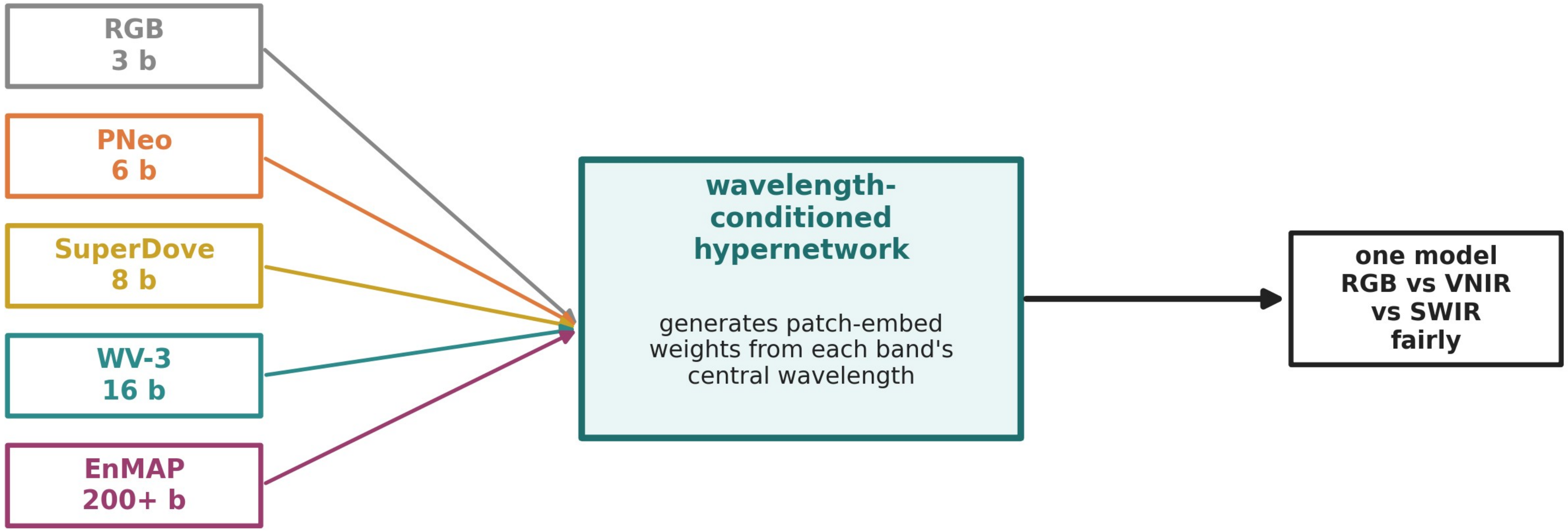
Method — a fair backbone

Foundation models for Earth Observation: a new lever

- 2024–2025: a wave of pretrained EO backbones (Prithvi, SpectralGPT, SoftCon, AnySat, DOFA…).
- Most are pretrained at 10–30 m on Sentinel-2 / HLS — naïve transfer to ~1.2 m VHR is not guaranteed.
- Two routes forward:
  - sensor-agnostic models (DOFA, AnySat) accept arbitrary band sets;
  - parameter-efficient adapters (DEFLECT) freeze the backbone, add $< 1\,\%$ params.
- Open question: does 10–30 m pretraining transfer to VHR? — empirical.

FM-RS survey 2024 · Xiong 2024 · Astruc 2024 (AnySat) · Thoreau 2025 (DEFLECT) · Szwarcman 2024 (Prithvi-EO-2.0)

DOFA: a band-agnostic backbone makes the comparison fair



Apples-to-apples, not apples-to-oranges: the same backbone ingests any band set keyed by wavelength.

Xiong et al. 2024 (DOFA, arXiv:2403.15356) — pretrained on 5 modalities, handles 202 bands

Xiong et al. 2024 (arXiv:2403.15356) — pretrained on 5 modalities (8M samples), handles up to 202 bands

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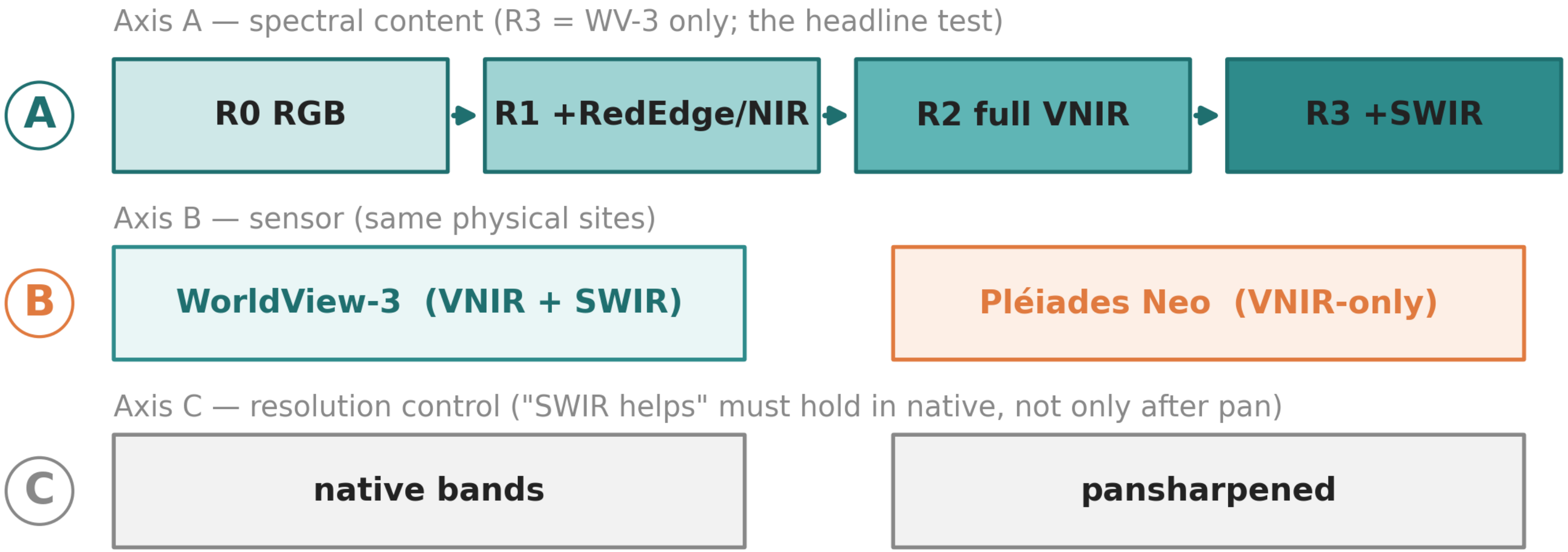
The gap & the proposed direction

Gaps in the literature — and the test that matters

- No public VHR dataset pairs waste sites with MATERIAL labels; no peer-reviewed Pléiades-Neo material benchmark exists.
- RGB pipelines conflate iso-colour, chemically distinct materials.
- No pipeline goes from spectra to a regulatory hazard class (material → EWC → risk tier).
- The object-vs-material split is unaddressed; generalizzazione (cross-region / sensor / time) is rarely shown.
- The test that matters: how much does multiband actually beat RGB for material — and where does it stop helping?

Gap list distilled from Fraternali et al. 2024 (arXiv:2402.09066)

The experiment: a 3-axis band ablation



Same DOFA backbone throughout · Swin-RSP = RGB reference · run per-pixel AND full-CNN → B−A = pure chemistry gain, D−C = total MS gain

Design: loop_prof_sota/04_experimental_design.md

Generalization as a first-class axis (expected pattern, to be measured)

	In-domain	Cross-region	Cross-sensor
R0 RGB	baseline	−5.1% (Gibellini)	ports trivially
R2 full VNIR	+ small	gap narrows	needs harmonization
R3 +SWIR	+ chemistry	gap narrows most	widens unless SBAF

Whether added bands narrow or widen the generalization gap is publishable either way — report ID−OOD ΔF1 per cell.

Cross-region −5.1%: Gibellini 2025 · cross-sensor: GeoCrossBench — cell values EXPECTED

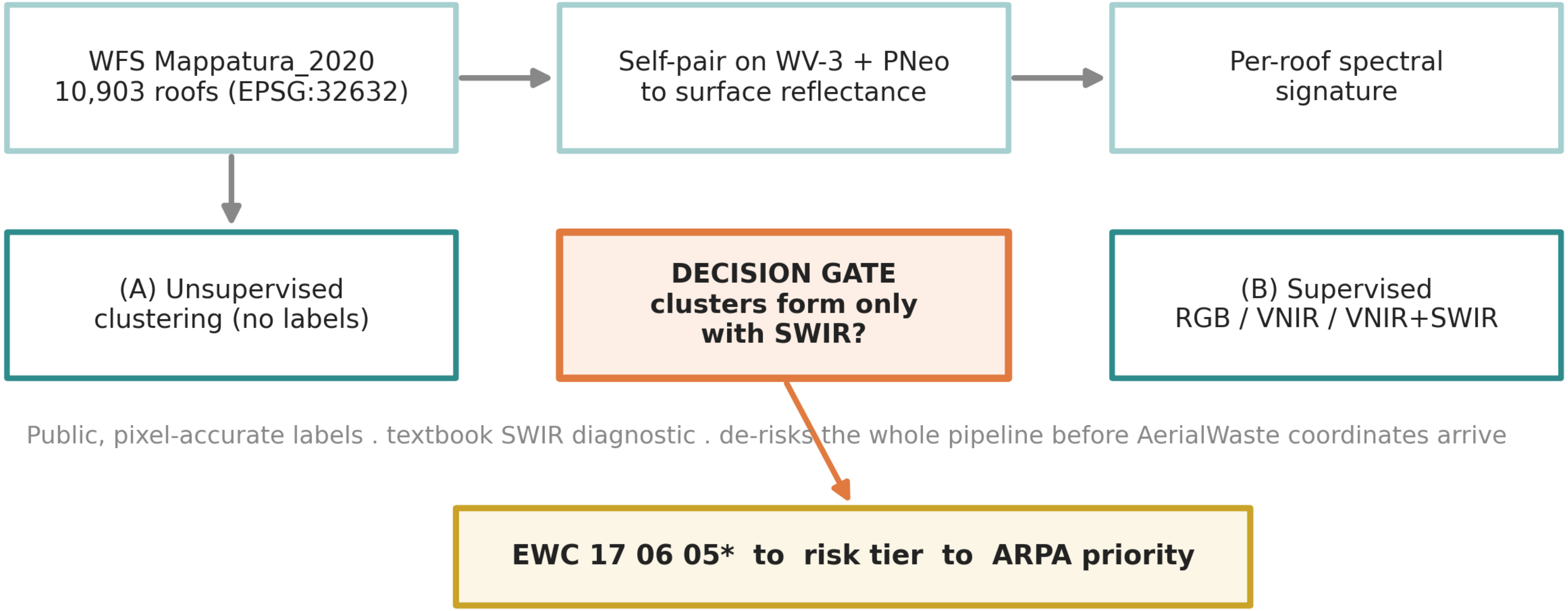
From material to risk tier

- Map predicted material → EWC hazard code → Indice di Degrado / PRAL logic × exposure × magnitude → ARPA priority tier.
- Output = a ranked intervention list; every step transparent and auditable.

  – Indice di Degrado (ID ≤25 / 26–44 / ≥45 → action windows) is NOT in the public WFS → estimated remotely (SWIR Mg-OH depth + VNIR weathering) = a thesis contribution.

  – Exposure grounded in Fazzo et al. (residential-proximity mesothelioma risk); magnitude from roof/site area.

Phase-1 asbestos pilot: the immediately-feasible demonstrator



GT: Lombardia WFS Mappatura_2020 . physics: chrysotile Mg-OH 2.30-2.33um (splib07a)

Public pixel-accurate labels + textbook SWIR diagnostic (Mg-OH 2.30–2.33 μm) — de-risks the whole pipeline

Why this matters — and what is genuinely novel

- First to MEASURE RGB vs VNIR vs SWIR added value for waste material at VHR.
- First WorldView-3 vs Pléiades Neo head-to-head for this task (SWIR vs VNIR-only).
- First to connect spectra → EWC hazard → ARPA risk tier — confidence-aware (reports where MS helps and where it doesn't, incl. generalizzazione).
  - Payoff: a transferable, free-data decision-support tool for environmental agencies.

Contributions map one-to-one onto the literature gaps

What to confirm with Thomas — decision checklist

- Data: trigger the ESA TPM proposal now (~9-week lead)? AOI / scene selection over Lombardia?
- Risk-chain depth: full Indice di Degrado / PRAL formalization, or stop at material + EWC?
- SuperDove: kept as a free wide-area screening layer, or dropped?
- Ablation depth: 3 axes as designed, or trim for time?
- Backbone: DOFA primary + Swin-RSP RGB reference — agreed?
- AerialWaste coordinate bridge with ARPA (gates the general-waste MS ablation).

Open dependencies flagged in loop_prof_sota/00_LOOP_LOG.md

A
Appendix — backup

Sensor comparison (full)

Sensor	VNIR	SWIR	Pan	Access	Role
WorldView-3	8 @ 1.24 m	8 @ 3.7 m	0.31 m	commercial / TPM	full ladder incl. R3
Pléiades Neo	6 @ 1.2 m	none	0.30 m	commercial / TPM	VNIR-only cross-sensor
SuperDove	8 @ 3 m	none	—	free, near-daily	wide-area screening
Sentinel-2	10–20 m	20 m	—	free	too coarse for material
EnMAP / PRISMA	HSI ~6.5 nm	yes	—	free (proposal)	30 m — context

Specs: Maxar · Airbus · Planet · ESA · ASI · DLR

The 13 target materials: shape vs chemistry

- Shape-identifiable (8 — RGB usually enough): vehicles, tanks, containers, rubble, bulky, wood, tyres, big-bags.
- Chemistry-bound (5 — RGB blind, MS lives here): asbestos, plastic-type, foundry slag, sludge, scrap-metal composition.
  – Asbestos + plastics are ideal test beds — shape cue AND chemistry cue coexist.

Taxonomy: loop_prof_sota/09_master_cheatsheet.md

Diagnostic feature → band (memorise)

Material	Feature (µm)	WV-3 band	Pléiades Neo?
Asbestos (Mg-OH)	2.30–2.33	SWIR-7 (2.33)	✗
Plastics (C-H)	1.215 / 1.73 / 2.31	S1, S4, S8	✗
Concrete / C&D	2.34 / 2.20	S8 / S6	✗
Slag / rust (Fe-oxide)	0.87–0.95	NIR1 / NIR2	weak
AC weathering	0.68 + 0.74	Red + RedEdge	✓

USGS splib07a · Aguilar 2021/2025 · Cilia 2015

Honesty guardrails & claims to handle with care

- Multiband value is measured, not assumed — and bounded (SWIR earns its keep on chemistry; detection may need little beyond RGB).
- Spectra at 1.24 m / 3.7 m vs pan 0.3 m = texture, not chemistry; SWIR carries chemistry but Pléiades Neo has none.
- Unmixing is out of scope (it sets the ceiling). “generalizzazione”, not “OOD”.

  – Downgraded / do-not-quote: Aguilar “+14 % κ” (use OA only); Saba “HS 97.3 vs MS 74.4” (paywalled); chrysotile “~5 nm” (it is tens of nm).

Key references

- Gibellini et al. 2025 — DL waste pipeline (Swin-T+RSP baseline). Waste Mgmt Bull.
- Fraternali et al. 2024 — waste RS survey / gap analysis. arXiv:2402.09066.
- Aguilar et al. 2021 — WV-3 VNIR+SWIR ablation. Remote Sensing 13(11):2133.
- Xiong et al. 2024 — DOFA band-agnostic foundation model. arXiv:2403.15356.
- Bonifazi 2026 (Geomatics) · Saba 2026 (J. Hazard. Mater.) · Shepherd 2025 (Sci. Rep.) — asbestos.
- Zhang & Ma 2024 — CascadeDumpNet (Pléiades). RSE. · Kokaly et al. 2017 — USGS splib07a.
- Full annotated bibliography: docs/02_research/loop_prof_sota/11_references.md + references.bib